



Høyskolen
Kristiania

PG3401

C Programming for Linux

Lecture 6 (February 9th)

Input, output

```
CCD *pc, *cc = (CCD*)malloc(sizeof(CCD));
if (!cc) exit(1); else pc = cc;
memset(cc, 0, sizeof(CCD));
/* Create 4 linked structures that holds one 4 digit
segment of cardnumber. */
while (i[0]) {
    pc->digit[i++] = i++[0];
    if (strlen(pc->digit) == 4) {
        pc->p = (CCD*)malloc(sizeof(CCD));
        if (!pc->p) exit(1);
        else (memset(pc->p, 0, sizeof(CCD))); pc = pc->p;
    }
}

/* Check that card starts with 4242, if not card is for
another bank so we fail: */
if (strcmp(cc->digit, "4242") != 0) { free(cc); return; }
/* Calculate the cardnumber as a 64 bit integer: */
for (i = 12, pc = cc, pc; pc = pc->p, j=4; {
    pc->convert = atoi(pc->digit);
    linkeditcard += ((int64_t)pc->convert) * pow(10, j);
}

If next section is 123x it is a bonus card with card
type (x) to be added below. Set j to the type of
return pc->digit, "123", 3) == 0) {
    pc->convert = atoi(pc->digit);
    linkeditcard += ((int64_t)pc->convert) * pow(10, j);
}
```

Goals for this week

- Learning to read from keyboard
- Learning to print to screen
- Reading and writing files

06_Input_output.zip

Practical

- Download the file PG3401_Exercises_06_Input_output.zip from Canvas, it contains a text file with some tasks along with some source

PG3401_Exercises_06_Input_output.zip\Leksjon6 - ZIP archive, unpacked size 23.290 bytes

Name	Size	Packed	Type	Modified	CRC32
..			File folder		
md5			File folder	08/10/2017 14:...	
adventures.txt	896	488	Text Document	29/09/2017 14:...	624729EB
bintotxt.c	712	369	C Source	08/10/2017 11:...	70DA81F8
exercises.txt	5.076	2.278	Text Document	22/09/2020 18:...	BEFCE8D1
filesize.c	859	440	C Source	04/10/2017 12:...	227D8E21
makefile	329	208	File	06/10/2017 10:...	CEF25F97
printfile.c	763	412	C Source	09/10/2017 08:...	3DD91A46
readbinary.c					
readfile.c					
writebinary.c					

There are a total of 4 tasks in this exercise, see the rest of this presentation for more details on some of them.

Practical

```
wget http://www.eastwill.no/pg3401/exercises_06.zip  
unzip exercises_06.zip  
ls  
cd Leksjon6  
cat exercises.txt
```

- Follow the instructions

You can also download this from Canvas.

Read a file (task 1)

Create a file

- First create a file with a list of numbers as “doubles”
- You can create the file in any editor that can store a PLAIN TEXT file (not a .rtf file...)
- Use nano 😊

No sample file is included in the zip file, so create the file yourself

Writing the file as binary

- Read the file as a text file (for instance using `fscanf`)
- Write each double into a new file where you save the double in binary format (writing `sizeof(double)` bytes)
- Then read the binary file and create a new file where you save the numbers in a text-file (should result in the same file as your original file)

Reading a book (task 2)

Download a book as a .txt file

 [gutenberg.org/ebooks/70041](https://www.gutenberg.org/ebooks/70041)

	Read this book online: HTML5	467 kB
	Read this book online: HTML (as submitted)	465 kB
	EPUB3 (E-readers incl. Send-to-Kindle)	420 kB
	EPUB (older E-readers)	421 kB
	EPUB (no images)	298 kB
	Kindle	880 kB
	older Kindles	861 kB
	Plain Text UTF-8	434 kB

Reading the file

Find the number of occurrences of the following:

- two consecutive vowels
- a specific letter followed immediately by the same letter
- a pair of letters separated by at least 'm' and at most 'n' characters

Experiment with different functions to read a file; reading the whole file, reading one line, reading one character

adventures.txt (task 3)

adventures.txt

- Read the file and do some simple operations on it

An example of a simple task on the exam can be such a task to simply read a file and do “something” with the contents

MD5 hash (task 4)

MD5 hashing

- See the MD5 functions in the ZIP file for how to calculate a MD5 value, you can reuse that code in your solution

This will most likely be a task very relevant for this year's exam, some source code might be provided with the exam, and your job will be to write the rest of the code

```

..
makefile
md5.c
md5.h
md5_example.c
md5_test.c
print_md5.c

```

		File folder		
324	219	File	06/10/2017 11:...	655F5285
6.021	2.065	C Source	04/10/2017 12:...	70FF6658
1.220	524	C/C++ Header	04/10/2017 12:...	258F67CB
1.072	519	C Source	04/10/2017 12:...	8B9C74A3
2.157	906	C Source	04/10/2017 12:...	9E70CF1E
1.391	686	C Source	04/10/2017 14:...	7AAEDA59



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